



Revisiting OpenStack and Big Data

Around five years ago, we explored Big Data using a desktop in *Open Source For You*. It is worth revisiting this article to see how the world has changed. You can find the old articles at <https://goo.gl/hPNLpg>.

Today, the broadband Internet is inexpensive and easily available. Hence, you are more likely to experiment using a free developer's account with one of the cloud services from Amazon, Google, or Microsoft. There are many more options as well, like Red Hat's OpenShift Online. You can find help and tutorials for getting started with each platform.

However, creating your own development environment on a local machine is still worth pursuing. It is a great learning experience.

In order to simulate a local cloud, I had used OpenStack to create and manage virtual machines on the desktop. OpenStack is an 'Infrastructure-as-a-Service' (IaaS) and is that part of OpenShift, which is classified as a 'Platform-as-a-Service' (PaaS).

I do not remember if I had heard of containers or Docker at the time. However, today, we are more likely to be using Docker containers instead of OpenStack for our experimentation on a local machine.

Docker

A container packages an application and its dependencies in an image. However, unlike a virtual machine (VM) image, it does not include the operating system. It creates a virtual environment (VE) for running the application on a common host. It is, thus, less expensive in terms of resources than a VM.

Docker has become the *de facto* option for containers as it makes it easy to package, manage and run the containers. Docker has an open source version and an enterprise version. The latter has additional, closed source functionality.

A Docker image is portable. Hence, you may use it for developing and testing on a local

machine, and install it on a private server or any cloud environment.

It is one of the newer technologies well worth learning as it is now widely used, and is especially useful when microservices are used in designing and implementing complex applications.

Kubernetes

Instead of multiple VMs, you now have multiple containers which will be run on a cluster of machines. For your own experimentation purposes at home, the cluster may be just a single well-configured desktop. So how do you conveniently deploy, manage and scale the containers on your cluster? The shorthand term for this activity has become 'orchestration' — obviously a metaphor borrowed from the music world.

Docker comes with its own orchestration tool, Swarm, which is easy to use with Docker containers.

However, a more complex but versatile tool is Kubernetes, which was developed and released as an open source product by Google. As you may expect, the motivation was to make the use of Google Cloud easier for people. The same tool can now be used to automate the running of containers across cloud computing from various vendors.

Hence, even though Kubernetes on a local machine is an overkill, it is well worth the effort to learn it.

Big Data products

I expected the Big Data world to have changed a lot in five years. But I was pleasantly surprised that the products we explored five years ago continue to be very significant even today.

Hadoop and HDFS are still the core

Hadoop MapReduce and Apache Spark continue to be the dominant Big Data analysis tools. Spark was